

MAKERERE UNIVERSITY BUSINESS SCHOOL
FINAL EXAMINATION FOR THE DEGREE OF
BACHELOR OF BUSINESS COMPUTING
OF MAKERERE UNIVERSITY ACADEMIC YEAR 2024/2025

COURSE NAME: COMPUTING MATHEMATICS

YEAR OF STUDY: ONE.

COURSE CODE: BUC 1114

SEMESTER: ONE

DATE: Tuesday 26th November 2024

TIME: 9.00am – 12.00pm

INSTRUCTIONS:

1. Time allowed is 3 hours
2. This examination contains two sections, A and B.
3. Section A is compulsory and carries 40 marks.
4. Section B carries 60 marks. Attempt **ONLY THREE** questions from Section B

Section A (40 Marks). Attempt all questions

1. Convert the binary number 1001.01012 to its decimal equivalent. (4marks)
2. Construct the truth table for the logical expression $P \vee \neg R$. (4 marks)
3. Given $\mu = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$, $V = \begin{bmatrix} -3 \\ 5 \\ -1 \end{bmatrix}$, find the values of
 - i) $\mu + V$ (2 Marks)
 - ii) $|\mu \cdot V|$ (2 Marks)
4. Given that Set A is the set of all counting numbers less than 10 while Set B is the set of even numbers less than 10. Explain whether the union of two sets A and B is the same as their intersection while giving evidence from this scenario (4 marks)
5. Evaluate the two's complement of -12 in 8-bit system (4marks)
6. Let R be a relation on the set $S = \{1, 2, 3, 4\}$ defined as follows: for any $a, b \in S$, $a R b$ if and only if $a \equiv b \pmod{2}$. Is R an equivalence relation on S? Justify your answer (4 marks).
7. List down four applications of linear algebra and matrices in computing. (4 marks)
 8. (a) Briefly explain the term algorithm in computer science. (2 marks)
 - (b) Distinguish between sequence and iteration algorithms. (2 marks)
9. The complexity of algorithms is largely governed by running time and space requirements. Outline the factors that influence:
 - a. time complexity (2 marks)
 - b. space complexity (2 marks)
10. Use a truth table to show that $P \rightarrow Q \equiv \neg P \vee Q$ (4 marks)

Section B (60 Marks). Attempt any THREE questions

Question 11

- a) Convert 101101002 to decimal using double dabble method. (4marks)
- b) Boeing 737 aircraft navigation systems rely on real-time calculations of its latitude, longitude, altitude, and speed. These values are continuously updated as the aircraft moves. The aircraft recorded its speed at 485.75 knots for a particular altitude. The

aircraft used IEEE 754 single precision floating point format to store its speed accurately within the range and precision needed for navigation updates in order to ensure safety and efficiency of the flight.

- i. Demonstrate the representation of the aircraft speed in the format mentioned above. **(8marks)**
- ii. Explain four importances of the format in part (i) above. **(8marks)**

Question 12:

a) A login system requires that both a username and a password must be correct for a user to gain access. If either the username or password is incorrect, access is denied. If the propositions are: U: "The username is correct." and P: "The password is correct.",

- i. Define a propositional logic expression to represent this system. **(2 marks)**
- ii. Draw a truth table for this system and show the cases where access will be denied. **(4 marks)**

b) In a computer system, there is a security feature that allows access to a specific file. The file can only be accessed under the following conditions:

- Condition 1: The user must be logged in (represented by L).
- Condition 2: The user must have the appropriate permissions (represented by P).

The file can be accessed if either of these conditions is true, or both.

- i. Write a logical expression to represent the access condition **(2 marks)**
- ii. Explain the implication of the above expression to the security system. **(4 marks)**

c) In an e-commerce system, you have two conditions:

- P: A customer has made a purchase.
- Q: The customer has provided their contact information.

The system should send a follow-up email only if the customer has made a purchase and provided their contact information. Using propositional logic,

- i. Write a logical expression that represents the condition for sending a follow-up email. **(2 marks)**
- ii. Determine whether the system will send the email if:
 - a. The customer has made a purchase but has not provided their contact information. **(2 marks)**
 - b. The customer has neither made a purchase nor provided their contact information. **(2 marks)**
 - c. The customer has made a purchase and provided their contact information. **(2 marks)**

Question 13:

A new ride-sharing company, Safeboda, uses a function to calculate the fare based on the distance travelled. For any trip, the fare $f(x)$ (in shillings) is calculated by the function $f(x)=4x+1500$, where x represents the distance in kilometres. Safeboda also offers a "Fare Estimate" tool that provides an inverse function, allowing users to calculate the estimated distance for a given fare.

Required;

- i) What would the fare be if the distance travelled is 500 kilometres? (4 marks)
- ii) Find the inverse of the function $f(x) = 4x + 1500$ and explain what it represents in terms of the fare and distance. (6 marks)
- iii) What distance would be travelled by a client who has shillings 5,000? (4 marks)
- iv) Explain any three (3) applications of functions in computing (6 marks)

✓ Question 14:

- a) Define the following
 - i. A matrix (1 mark)
 - ii. Order of a matrix (1 mark)
 - iii. Identity matrix (1 mark)
- b) Given that $A = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}$, determine;
 - i) Determinant of matrix A (2 marks)
 - ii) A^{-1} (2 marks)
 - iii) A^T (2 marks)
- c) Find the values of x and y in each of the following;
 - i) $\begin{pmatrix} 3 & -5 \\ 2 & x \end{pmatrix} + \begin{pmatrix} 1 & y \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 5 & -2 \end{pmatrix}$ (2 marks)
- d) Systems of linear equations can be solved using various methods
 - i. Describe any four methods of solving systems of linear equations. (4 marks)
 - ii. Using a method of your choice, solve the following systems of linear equations (5 marks)

$$2x - y = 1$$

$$x + 3y = 4$$

✓ Question 15:

- a) Briefly explain five properties of algorithms. (5 marks)
- b) Kirabo suspects that she could have contracted malaria sickness. She lives 10 km from the nearest hospital. The hospital has doctors, a pharmacy and a laboratory. Write a simple algorithm, in natural language of the processes that Kirabo has to undertake so as to address her challenge. (4 marks)
- c) Construct a flow chart for the algorithm in (b) above (5 marks)
- d) Explain three applications of algorithms in computer science. Give an example in each case. (6 marks)

End of Question paper.